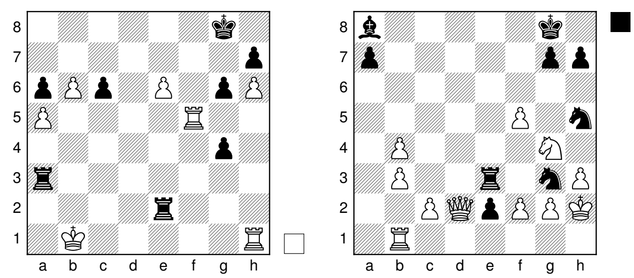
Conditional Generation of Creative Chess Puzzles with Diffusion Models

Anonymous submission

Abstract

While modern language models demonstrate impressive generative capabilities, they often struggle with constrained, counter-intuitive creative tasks. To address this limitation, we explore chess puzzle generation as a rigorous testbed for computational creativity and reasoning, a domain where altering a single piece can invalidate an entire solution. We propose a novel approach for conditional generation of creative chess puzzles using masked diffusion models. Unlike previous methods, our non-directional diffusion approach allows for conditioning on specific tactical themes and partial board positions. We introduce a novel auxiliary task of simultaneous best-move prediction, which improves solution uniqueness by 11.8%, theme-conditioning accuracy by 2.4%, and puzzle rate by 15.8%. To further optimize solution uniqueness and counter-intuitiveness, we establish a reinforcement learning framework adapted from Denoising Diffusion Policy Optimization (DDPO). This RL phase increases the yield of unique, counter-intuitive puzzles by 97.7%. Finally, we release the first open-weights models for chess puzzle generation, offering a new pathway for controllable, creative generation.



(a) *Themes:* advantage, short, endgame, rook endgame.
Solution: Rb5 cxb5 b7

(b) *Themes:* fork, under promotion, promotion, endgame
Solution: Nf1+ Rxf1 exf1=N+ Kg1 Nxd2

Figure 1: Example chess puzzles generated by the proposed model. The themes are provided as input to the model and the solutions are shown below each puzzle.

Introduction

The idea of machine intelligence has intrigued researchers for decades, dating back to Turing's early work on indistinguishable machine behavior (Turing 1950). In recent years, this pursuit has culminated in the rapid rise of Large Language Models (LLMs), which might already be capable of passing standard Turing test (Jones and Bergen 2026). A key benchmark for evaluating the success of these systems is creativity (Colton and Wiggins 2012). However, while modern language models are capable of creative work, such as creating art, writing stories or making lyrics, they struggle to come up with truly new and unique items (Wenger and Kenett 2026). To address this limitation, we attempt to enhance the controllability of creative generation with a small, domain-specific language model.

While the most commonly used language models today are autoregressive, discrete diffusion models have recently been successfully adapted to discrete sequences and language modeling tasks (Austin et al. 2021; Nie et al. 2025b; Ye et al. 2025b). Autoregressive models have generally been outperforming diffusion models with respect to output quality, but recent works demonstrate competitive results using diffusion models (Nie et al. 2025b; Ye et al. 2025b). Moreover, they can yield significant speedups, offering compelling quality–inference speed tradeoffs enabled by parallel decoding (Wu et al. 2026). While autoregressive generation builds sequences token-by-token, diffusion models iteratively refine the entire sequence globally (Shi et al. 2024). This non-directional generation process offers a promising alternative for creative generation, and hence we select a masked diffusion model for our task.

Evaluating the creative abilities of language models is an open question, and previous work has turned to chess puzzle generation as a contained playground (Feng et al. 2026). We employ this same framework as it allows us to concentrate on the controllability of creative generation. This domain requires the models to balance between creativity and reasoning, as chess puzzles are generally delicate. Changing even one piece on the board may entirely change the puzzle to something completely different.

To train our model for this task, we first employ supervised training on the Lichess puzzles dataset (Duplessis 2026b). Our model takes a list of themes as input and generates positions, which match the given themes. After supervised training, we observe that around 0.44% of the positions generated by our model are valid puzzles, defined as chess positions which have a unique best solution and which are sufficiently counter-intuitive, following Feng et al. (2026). After supervised training, we employ reinforcement learning, where we use the puzzle criteria as the reward function, allowing us to nearly double the puzzle rate to 0.87%. Note that even in the Lichess training dataset, only 2.13% pass the puzzle criteria. Moreover, given the small model size (268M parameters), we can efficiently generate a large number of puzzles and filter out the ones not passing the criteria.

The main contributions of this work are the following:

1. We demonstrate the effectiveness of masked diffusion models for creative chess puzzle generation. In contrast to previous work that has primarily focused on autoregressive models (Feng et al. 2026), the proposed diffusion model provides the user a significantly greater control over the creative process by supporting conditioning of the puzzle generation on (i) a set of themes, (ii) a partial position, and (iii) the best move. This unlocks new practical possibilities such as practicing specific types of puzzles that a user wants to improve on.
2. We propose a novel auxiliary task of predicting the best move together with generating the puzzle. We show that this improves the solution uniqueness, accuracy of theme-conditioning and puzzle rate by 11.8%, 2.4% and 15.8%, respectively.
3. We introduce an effective RL recipe on top of masked diffusion models, which increases the rate of unique and counter-intuitive puzzles by 97.7%. Getting RL to work effectively is challenging with language models in general and it is less explored on top of discrete diffusion models.
4. We release the first open weights models for chess puzzle generation.

Example puzzles generated by our model are presented in Figure 1. For each puzzle, the solution and the themes on which we conditioned the generation are in the caption. These puzzles were hand-picked from the generated puzzles based on the elegance of the solution.

Related work

Machine intelligence and creativity Evaluating and expanding machine intelligence through computational creativity has long served as a key benchmark for artificial intelligence systems (Turing 1950; Colton and Wiggins 2012). While modern Generative AI models demonstrate strong capabilities across unstructured tasks like text and image generation, they often exhibit limited capacity for open-ended discovery or generating truly novel, highly constrained, counter-intuitive artifacts (Wenger and Kenett 2026). To evaluate and improve generative models beyond standard tasks, researchers increasingly investigate specialized domains requiring both aesthetic quality and tight computational constraints.

To address these limitations, recent work explores pairing generative architectures with evolutionary search and expert evaluation in specialized domains. For instance, Earle et al. (2026) substituted human selectors with vision-language models (VLMs) in the canonical Picbreeder framework to test open-ended semantic discovery, while Banarse et al. (2026) integrated genetic algorithms with foundation models to co-evolve complex 3D forms. Beyond purely visual media, AI pipelines have been applied to constrained spatial tasks like flat-foldable origami design (Zahavy, Hou et al. 2026), and recent empirical evaluations confirm that expert reviewers frequently rate machine-generated creative outputs as highly novel and counter-intuitive compared to human baselines (Veeriah et al. 2025).

Chess puzzle generation Although chess puzzle generation is a niche research field, it has been studied quite extensively despite its limited practical applicability. Traditional ways of generating chess puzzles include evolutionary search approaches, as well as iterative approaches. These algorithms iteratively modify one or a set of chess positions with a goal of making better puzzles (Feng et al. 2026; Iqbal 2021, 2017). Both algorithms use a fitness function that assesses qualities similar to our reward function (6). One more modern approach is to use deep learning for this task (Feng et al. 2026). Feng et al. train a model that generates chess puzzles. They manage to train a model that successfully beats a baseline of the Lichess puzzles dataset (Duplessis 2026b) in terms of counter-intuitivity. We do not position our work as attempting to beat their baseline in puzzle quality metrics. Instead, we attempt to add additional features useful for actual use, such as conditional generation. Additionally, an open source implementation allows anyone to use our model in practice.

Masked diffusion models While autoregressive large language models (LLMs) currently dominate language modelling, discrete diffusion models have emerged as a compelling alternative (Nie et al. 2025b; Ye et al. 2025b). Unlike autoregressive generation, which generates sequences token-by-token in a causal manner, discrete diffusion models iteratively refine the entire sequence globally using bidirectional attention (Austin et al. 2021). This non-causal structure allows the model to leverage context from the entire input sequence simultaneously and has been shown to reduce training data memorization (Luo et al. 2026). Furthermore, diffusion models offer variable inference-time efficiency by adjusting the number of denoising steps (Austin et al. 2021). However, they are computationally more expensive to pretrain to achieve comparable text quality (Nie et al. 2025a). Some tasks, which do not have the inherent left-to-right structure of language, such as sudoku might even benefit from the approach of diffusion models, where the order of the generated tokens is not fixed (Ye et al. 2025a).

Reinforcement learning with diffusion models In recent years, great effort has been put into improving reinforcement learning for diffusion models (Ou et al. 2026; Zhao et al. 2026; Wang et al. 2026; Rojas et al. 2026; Black et al. 2024; Fan et al. 2023). Compared to traditional LLMs, diffusion models suffer from computationally intractable sequence-level likelihoods. More traditional methods solve this by evaluating the likelihoods step-wise, which requires many forward passes (Black et al. 2024; Fan et al. 2023), whereas other algorithms approximate the likelihoods with one forward pass (Ou et al. 2026; Zhao et al. 2026; Wang et al. 2026;

Rojas et al. 2026). In this research, we initially attempted to use the ELBO-based sequence-level policy optimization (ESPO) algorithm (Ou et al. 2026), however, we were unable to find a working configuration. Afterwards, we turned to denoising diffusion policy optimization (DDPO), which we found to perform better despite being less efficient in theory.

Methods

Masked diffusion models

We frame our approach within the masked diffusion paradigm, where sequences are generated by iteratively choosing tokens from a vocabulary to replace mask tokens [MASK] (Shi et al. 2024). The forward process corrupts an unmasked sequence y into a noisy state y_t at time $t \in [0, 1]$ via $q(y_t \mid y) = \text{Cat}(y_t; \bar{Q}(t)^\top y)$. Using a masking schedule α_t, the transition matrix $\bar{Q}(t) = \alpha_t I + (1 - \alpha_t)\mathbf{1}e_m^\top$ transitions sequences from a fully unmasked to a completely masked as time increases (Shi et al. 2024).

To reverse this corruption, a denoising network $\mu_\theta(y_t, x)$ is introduced. The transition to a less masked state y_s ($s < t$) is defined as:

$$p_\theta(y_s \mid y_t, x) = \text{Cat}(y_s; \bar{R}^{\mu_\theta(y_t,x)}(t, s)^\top y_t), \tag{1}$$

where $\bar{R}$ is the reverse transition matrix (Shi et al. 2024). Conceptually, a token masked at step t is unmasked at step s with probability $\frac{\alpha_s - \alpha_t}{1 - \alpha_t}$, using a token sampled from the network's output. To represent the board in a machine-readable format, we pad the Forsyth-Edwards Notation (FEN) to a fixed length and convert it into tokens. Other implementation details are provided in Table A1 in the Appendix.

Supervised training

During supervised training, we optimize the evidence lower bound (ELBO) on the Lichess puzzle dataset (Shi et al. 2024):

$$\begin{aligned}
&\mathcal{L}_\theta(y \mid x) = \\
&\quad -\int_0^1 \frac{\alpha_t'}{1 - \alpha_t} \mathbb{E}_{q(y_t|y)} \Bigg[\\
&\quad \sum_{i=1}^{L} 1[y_t^{(i)} = [\texttt{MASK}]] \cdot (y^{(i)})^\top \log \mu_\theta(y_t, x)^{(i)} \Bigg] \mathrm{d}t,
\end{aligned} \tag{2}$$

where L is the sequence length, x is the conditioning information, and $y_t \sim q(y_t \mid y)$ is the corrupted sequence. Because calculating the exact integral and expectation is computationally intractable, we employ a 1-point Monte-Carlo (MC) estimate. A comprehensive list of all hyperparameters used during training can be found in Table A1 in the Appendix. Following previous methodology, we select the checkpoint with the lowest validation loss, which occurs at 1,000,000 steps (Feng et al. 2026).

To evaluate how best move prediction affects model performance, we train two different configurations: one that generates both the puzzle and the best move, and another that only generates the puzzle. In addition to these configurations, we also experiment with block diffusion to investigate the effect of best move prediction on puzzle quality. With block diffusion, we split the generative process into two phases: the first phase generates the board state while artificially keeping the best move tokens masked (Arriola et al. 2025). Only after all board tokens have been generated is the best move sampled (Arriola et al. 2025). We use this as a middle ground between the two standard models, as the block diffusion model has learned internal representations for the best move but is restricted from explicitly unmasking those tokens during board generation.

Denoising Diffusion Policy Optimization

Similarly to (Feng et al. 2026), we apply reinforcement learning to optimize the pretrained model for puzzle quality metrics. For RL training, we adapt the Denoising Diffusion Policy Optimization (DDPO) algorithm (Black et al. 2024). Although DDPO was originally developed for image-generation models, we adapt it to masked diffusion models by using the transition probabilities from the reverse process (1) as the policy $\pi_\theta(a_t \mid s_t, x) = p_\theta(a_t \mid s_t, x)$. Additionally, we adapt the group relative advantage calculation from GRPO (Shao et al. 2024), which is originally not part of DDPO. DDPO acts in a step-wise Markov Decision Process (MDP), where the states are defined to be a partially masked token sequences s_t. The actions a_t correspond to sampling a new partially masked token sequence with the transitions being deterministic. The reward is given at the end of the trajectory for the full generated sequence and propagated through the trajectory via discounting.

As shown by (Feng et al. 2026), a pure reward maximization procedure suffers from entropy collapse. Like their work, we employ diversity filtering mechanisms alongside an entropy bonus and a KL divergence penalty to prevent entropy collapse and ensure the realism of the puzzles. However, we optimize these penalties directly via gradient descent by incorporating them into the loss function. Using this setup, we seek to maximize the following objective:

$$\mathcal{J}(\theta) = L(\theta) - \beta \mathbb{E}_t \left[\mathbb{KL}(\pi_\theta \parallel \pi_{\text{ref}}) \right] + \gamma \mathbb{E}_t \left[H(\pi_\theta) \right] \tag{3}$$

where $L(\theta) = \mathbb{E}_t[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon)\hat{A}_t)]$ is the proximal policy optimization (PPO) objective (Schulman et al. 2017), and $\mathbb{E}_t \left[\mathbb{KL}(\pi_\theta \parallel \pi_{\text{ref}}) \right]$ and $\mathbb{E}_t \left[H(\pi_\theta) \right]$ are the expected step-wise KL divergence and the entropy, respectively. The advantages $\hat{A}_t$ are computed and normalized group-wise using an unbiased Monte Carlo estimate proposed by (Kool, van Hoof, and Welling 2019).

Detecting chess puzzles

Following Feng et al. (2026), we define a chess position to be a puzzle if it passes legality, uniqueness (4), and counterintuitiveness criteria (5). We attempt to define and implement the metrics in the same way as Feng et al. (2026), but minor unknown differences remain in the distance calculations.

Uniqueness A position s is unique if the winning chances after the best move are considerably larger than those after the second best move (Duplessis 2020; Feng et al. 2026).

$$\mathbb{I}_{\text{uni}}(s) = w(a_{\text{best}} \mid s) - w(a_{\text{second}} \mid s) > \tau_{\text{uni}}, \tag{4}$$

where $w(a \mid s)$ yields the winning probability of move a in position s (Duplessis 2026a) and $\tau_{\text{uni}} = 0.5$.

Counter-intuitiveness The counter-intuitiveness value of a position is defined as follows (Feng et al. 2026):

$$r_{\text{cnt}}(s) = 0.8 \cdot \frac{d_{\text{cp}}}{50} - 0.1 \cdot \frac{v_{\text{captured}}}{9} \quad (5)$$

where $d_{\text{cp}} \leq 50$ is the critical depth at which Stockfish first discovers the best move, and v_{captured} is the value of the potentially captured piece. A position is considered counte-intuitive if $r_{\text{cnt}}(s) > \tau_{\text{cnt}} = 0.1$.

Reward function We defined the RL reward function $R(s)$ as:

$$R(s) = \begin{cases} -2 & \text{if } \neg\mathbb{I}_{\text{legal}}(s) \\ \max(10 \cdot r_{\text{cnt}}(s), 0) & \text{if } \mathbb{I}_{\text{legal}}(s) \cdot \mathbb{I}_{\text{filter}}(s) \cdot \mathbb{I}_{\text{uni}}(s) \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

where $\mathbb{I}_{\text{filter}}(s) = \mathbb{I}_{\text{diversity}}(s) \cdot \mathbb{I}_{\text{pieces}}(s) \cdot \mathbb{I}_{\text{theme}}(s)$. $\mathbb{I}_{\text{pieces}}(s)$ filters out positions in which the piece count of some piece exceeds that of the starting position, and $\mathbb{I}_{\text{theme}}(s)$ indicates that the generated puzzle adheres the requested themes. The diversity indicator factorizes as $\mathbb{I}_{\text{diversity}}(s) = \mathbb{I}_{\text{entropy}}(s) \cdot \mathbb{I}_{\text{intra}}(s) \cdot \mathbb{I}_{\text{inter}}(s)$, where $\mathbb{I}_{\text{entropy}}(s)$ filters out low-entropy generations, $\mathbb{I}_{\text{intra}}(s)$ filters out positions that are too similar to each other in the same batch and $\mathbb{I}_{\text{inter}}(s)$ filters out positions that are too similar to positions in previous batches. Both intra- and inter-batch distances are computed using the Levenshtein distance (Yujian and Bo 2007) over the FEN and the principal variation (PV), where the PV distance is computed over the closest position in terms of the FEN distance. A position passes the conditions if the minimum distance to all other positions is greater than a threshold. For more details, we direct the reader to (Feng et al. 2026), as apart from the theme condition, the PV distance and the exact form of the reward function, our diversity filtering setup is very similar to theirs.

Experimental evaluation

Our experimental evaluation is structured as follows. We first analyze how best move prediction affects the puzzle generation process. We also compare our model to the ones trained by Feng et al. (2026). Second, we evaluate the reinforcement learning phase. Finally, we present how conditioning affects the generation, specifically during the RL phase.

Best move prediction improves puzzle metrics

Table 1 summarizes the puzzle metrics of the models trained with supervised training. Interestingly, we find that simultaneously generating the best move alongside the board state leads to measurable improvements in both uniqueness and theme match rates compared to generating the board alone or predicting the best move with block diffusion. Although the confidence intervals for the counter-intuitiveness and puzzle rates overlap across configurations, the model with joint best-move generation achieves the highest estimate for the final puzzle rate (0.18%). Across all supervised configurations, the theme match rate remains remarkably high, exceeding 84% for positions with unique solutions. This indicates that the model successfully aligns its generations with the requested conditioning in the vast majority of cases, reflecting a solid grasp of varied tactical concepts.

To quantify the novelty and diversity of the generated positions, we utilize the distance metrics summarized in Table 2. For the baseline Lichess puzzle dataset, the board distance is 11.88, which corresponds to a difference of roughly 12 distinct board elements between a puzzle and the most similar other position. The Lichess dataset exhibits a baseline PV distance of 0.95, representing a high diversity of solutions where the start or target squares of the moves typically differ by at least one rank or file. As shown in Table 2, all three supervised configurations closely preserve these distances. The board self-distances are only slightly lower than that of the original Lichess dataset, while the PV self-distance remains also quite similar. Furthermore, the board and PV distances to the Lichess dataset itself are high are also similar, which indicates that the distribution of generated positions is similar to the lichess dataset distribution. Importantly, the choice of the best-move generation strategy has virtually no impact on these diversity metrics, demonstrating that coupling chess understanding with the generation process to improve puzzle quality does not compromise diversity.

Comparing our model to the one trained by (Feng et al. 2026) in Table 1, we observe that after supervised training, our models have higher uniqueness and puzzle rates. In contrast, their model has similar legal rates as ours after RL. Additionally, their counter-intuitiveness is slightly higher than ours. It is easy, however, to increase the puzzle metrics at the cost of diversity. In Table 2, we observe that both board distances are considerably higher for our models compared to theirs. Notably, we have not managed to align the PV distance implementation to theirs. Therefore, the PV distances between our models and theirs are not comparable. However, overall, based on the board distance and the puzzle metrics, conditioning the generation on themes helps the model perform better in practice.

Reinforcement learning

For reinforcement learning, we initialize the policy from the supervised checkpoint trained with joint best-move prediction, as it achieved the highest baseline puzzle rate (0.44%). We fine-tune this model using the DDPO framework, incorporating the diversity mechanisms we described previously.

In Figure 2, we present the progression of our RL run. Figures 2a-2g showcase the evolution of the key performance metrics, whereas the rest of the plots concern the diversity of the generated positions. We observe that the initial improvement in the reward comes mainly from the improved legal rate. During training, the proportion of illegal positions decreases by an order of magnitude from around 1.0% to around 0.1%. The second metric that is improved is the counter-intuitive rate. The counter-intuitive rate for unique positions approximately doubles from under 1% to over 2%. The increased counter-intuitive rate also yields improvements in the puzzle rate, which improves from around 0.4% to over 0.8%. Lastly, we see a small decrease in the theme match rate and a small increase in the move match rate, despite not explicitly training for the best move.

For most of the RL training, the uniqueness rate stays

Table 1: Puzzle quality metrics for different model configurations. All models were sampled with a temperature 1.0 and 256 denoising steps. The errors represent the 95% standard errors.

Training	Best move	Legal (%)	Unique (%)	Counter-Intuitive (%)	Puzzle (%)	Theme (%)
Supervised	No	98.97 ± 0.02	39.14 ± 0.10	0.98 ± 0.03	0.38 ± 0.01	84.23 ± 0.08
Supervised	Block	98.97 ± 0.04	41.31 ± 0.22	0.94 ± 0.07	0.39 ± 0.03	85.53 ± 0.18
Supervised	Yes	98.95 ± 0.05	**43.76 ± 0.22**	1.00 ± 0.07	0.44 ± 0.03	**86.27 ± 0.19**
RL	Yes	**99.87 ± 0.01**	39.10 ± 0.10	**2.23 ± 0.05**	**0.87 ± 0.02**	78.72 ± 0.08
Supervised (Feng et al. 2026)		99.72	30.89	1.11	0.34	—
Lichess puzzles dataset		100.00 ± 0.00	94.44 ± 0.10	2.26 ± 0.06	2.13 ± 0.06	100.00 ± 0.00

Table 2: Self-distance and Lichess-distance metrics after supervised training and RL. Metrics are computed over positions passing the uniqueness filter. We align the number of positions with (Feng et al. 2026) and use 100k samples for Lichess-distances and 40k samples for the self-distances. Despite this, only board distances are comparable to (Feng et al. 2026) due to differing implementations of the PV distance.

Training	Best move	Board (Self)	Board (Lichess)	PV (Self)	PV (Lichess)
Supervised	No	11.66 ± 0.01	11.18 ± 0.01	0.78 ± 0.002	0.93 ± 0.001
Supervised	Block	11.67 ± 0.01	11.18 ± 0.01	0.79 ± 0.002	0.93 ± 0.001
Supervised	Yes	11.75 ± 0.01	11.24 ± 0.01	0.80 ± 0.002	0.93 ± 0.001
RL	Yes	15.15 ± 0.01	14.34 ± 0.01	0.76 ± 0.002	0.97 ± 0.001
Supervised (Feng et al. 2026)		8.528	10.859	(0.837)	(0.637)
Lichess puzzles dataset		11.80 ± 0.01	—	0.95 ± 0.001	—

approximately constant, however, after 4,500 training steps, the uniqueness rate rapidly increases. This sudden increase, however, is not a result of model getting better, but instead results from model collapse. As seen in Figures 2i-2k, the distance metrics initially increase, but after around 3,000 training steps, start to decrease. As novelty is an important success metric for our model, we select the checkpoint at 3,500 steps, which still has high diversity. Interestingly, the PV distance to the Lichess dataset in 2l never starts to decrease suggesting that the model has found a position, which yields high rewards, but is not present in the Lichess dataset. In fact, when manually observing the generated puzzles after the model collapse, we see that the model begins to generate only a few different positions. We present examples of such positions in Figure A3 in the Appendix. Interestingly, two such positions are not immediately ruled out by the current diversity filtering procedure, as the exact positioning of the White queen is not important for the solution in either position.

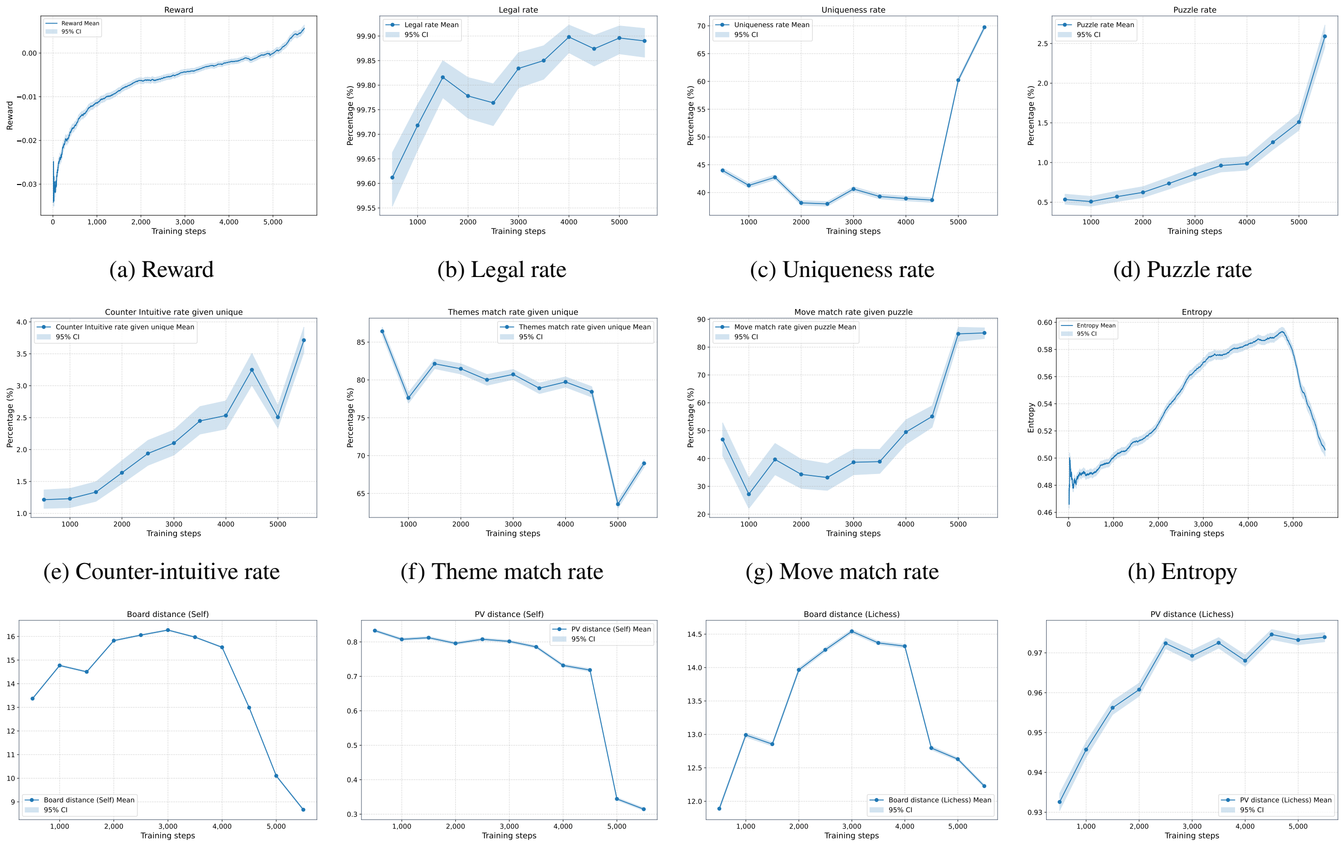
Table 1 presents the final puzzle metrics for the RL model at step 3,500. As seen with the training curves, we observe improvements in the legal and counter-intuitiveness rate, while see a slight decrease in uniqueness and theme match rates. Notably, although the uniqueness rate is slightly lower than in the base model, the puzzle rate is over twice as high, which is driven by the counter-intuitiveness rate. This observation is consistent with (Feng et al. 2026). In Table 2, we see that the RL training improved the novelty of the generated puzzles. While the PV distances remain approximately constant, the board distance metrics improve significantly. Nonetheless, the overall yield of unique, counter-intuitive, and theme-compliant puzzles increases from 0.16% to 0.32%.

Manual control via conditioning

A key contribution of our conditional generation approach is that it enables manual control over the generated puzzles. While unconditioned models, such as that of Feng et al. (2026), generate creative puzzles at random, they do not allow users to specify which tactics they want to practice. By conditioning our model on specific themes, we make the generator practically more useful, allowing users to generate targeted practice material for positions they struggle with.

Furthermore, conditioning the generation process is crucial for retaining the full diversity of chess tactics post-reinforcement learning. Because short or simple tactics (such as *one-move* or *mate-in-one* puzzles) are almost never counter-intuitive, a global reward naturally suppresses them in favor of longer, more complex configurations. In contrast, our RL setup enforces theme matching via the reward function (Equation 6), ensuring that the model is only rewarded if the generated puzzle satisfies the requested conditioning. Additionally, the advantages are computed over groups of same themes, which forces the policy to maximize puzzle quality within the context of the requested theme, rather than globally converging to the more difficult themes.

Figure 3 illustrates the distribution of tactical themes across the original Lichess dataset, our supervised baseline, and the final RL model. Both of our models successfully preserve the full spectrum of themes. In particular, themes like *mate-in-one* are successfully retained, despite the fact



(a) Reward (b) Legal rate (c) Uniqueness rate (d) Puzzle rate

(e) Counter-intuitive rate (f) Theme match rate (g) Move match rate (h) Entropy

(i) Board distance (Self) (j) PV distance (Self) (k) Board distance (Lichess) (l) PV distance (Lichess)

Figure 2: Training progress during reinforcement learning. Model collapse occurs after approximately 4,500 steps, characterized by a sudden drop in distances. Shaded areas represent 95% standard errors. For the distance metrics, we compute the self distances with only 10,000 positions.

that they are practically never counter-intuitive and would otherwise receive zero reward.

A key advantage of masked diffusion models compared to autoregressive models is that generation can be done in any order. This property allows one to condition the generative process on any partial token sequences, where only some parts of the sequence are masked. In particular, for our use case, we can select some parts of the board which we want to be present in the final puzzle and generate the remaining board. This would not be possible with autoregressive models natively, as the order of generation is decided beforehand.

In Figure 4 we present an example of conditioning on a partial board configuration and the best move. The initial configuration we use for conditioning is presented in Figure 4a. The best move is also used for conditioning, forcing the sacrifice of the bishop on h7 to be the best move. We position the black king in the typical position after castling king side. The remaining position is generated by the model. Figures 4b–4d present three different completions from the initial masked position. Although the first move of the solution is the same for each position, the continuation changes depending on the puzzle.

Conclusions and discussion

In this work, we explored the use of masked diffusion models for the generation of chess puzzles. We presented a novel approach that conditions the generative process on specific tactical themes, allowing users to practice targeted problems while maintaining high diversity across both simple and complex concepts. Additionally, we introduced the auxiliary task of simultaneous best-move prediction, which measurably enhances the model's chess understanding and overall generative capabilities. After supervised training, our model successfully generates valid puzzles with correct themes approximately 0.16% of the time. By applying our reinforcement learning framework adapted from Denoising Diffusion Policy Optimization, we were able to more than double this success rate to 0.32% while improving the counter-intuitiveness and diversity of the generated positions. Lastly, we release the first open-weights models for chess puzzle generation to the community.

Despite these promising results, we encountered several challenges and limitations during this research. Most notably, the reinforcement learning phase suffers from model collapse during extended training. After a certain point, the

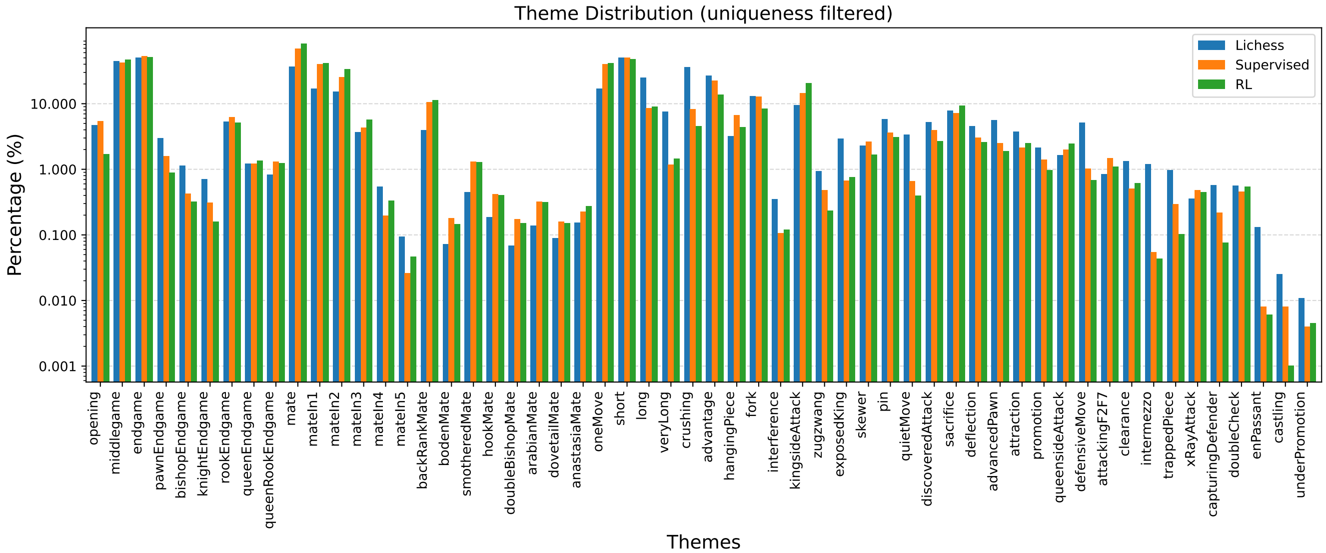
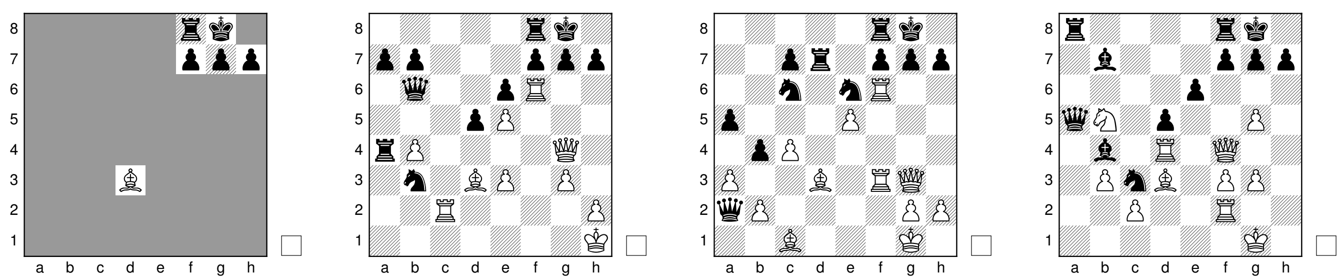


Figure 3: The distribution of themes for the Lichess puzzles dataset, our supervised model and the final RL model. All positions are first filtered for positions with a unique solutions and the percentages are computed across the filtered dataset.



(a) Starting position

(b) *Solution:* Bxh7+ Kxh7 Rc8 Qxb4 Qh5+ Kg8 Qxf7+ Kh7 Qg6+ Kg8 Rfxf8+ Qxf8 Qxe6+

(c) *Solution:* Bxh7+ Kxh7 Qh4+ Kg8 Rh3 Rd1+ Kf2 Rf1+ Kxf1 Qxc4+ Qxc4

(d) *Solution:* Bxh7+ Kxh7 Rh2+ Kg6 Rh6+ gxh6 Qf6+

Figure 4: Three positions generated from a common starting point. The only theme sampling was conditioned with is *sacrifice*. The generation was also conditioned on the bishop sacrifice Bxh7+ and that it is white's move next.

model converges on exploiting specific high-reward configurations, which compromises the diversity of generated positions achieved in earlier checkpoints. Furthermore, we observed a slight decrease in the theme match rate during reinforcement learning. This likely stems from the model discovering that certain theme combinations inherently make generating high-reward puzzles easier, causing the policy to gravitate toward these positions rather than accurately capturing rare theme combinations. Lastly, while our findings in this specific domain are robust, we have not yet demonstrated the generalization of these contributions to broader natural language processing tasks.

These limitations highlight clear avenues for future research. One possible direction is on further stabilizing the reinforcement learning process for discrete diffusion models. Refining the diversity filtering mechanisms, conditioning on more diverse sets of themes, or exploring alternative reward formulations could prevent entropy collapse and allow for longer, more stable training runs. Especially, using random theme combinations that are not present in the puzzles dataset in RL could improve diversity and increase the amount of positions interesting to humans. Furthermore, expanding the conditioning framework to define and train on much more sophisticated or abstract themes, such as king-on-tour or perpetual check, could yield even richer puzzle compositions. For new approaches to puzzle generation, combining iterative algorithms and deep learning architectures within a chain-of-thought-type system could offer additional benefits. Finally, applying this setup to other domains to showcase generalization would significantly advance the research boundary. The success of masked diffusion and auxiliary prediction in a domain as unforgiving as chess suggests this architecture could serve as a strong blueprint for other highly constrained generation tasks.

References

Arriola, M.; Sahoo, S. S.; Gokaslan, A.; Yang, Z.; Qi, Z.; Han, J.; Chiu, J. T.; and Kuleshov, V. 2025. Block Diffusion: Interpolating Between Autoregressive and Diffusion Language Models. In *The Thirteenth International Conference on Learning Representations*.

Austin, J.; Johnson, D. D.; Ho, J.; Tarlow, D.; and van den Berg, R. 2021. Structured Denoising Diffusion Models in Discrete State-Spaces. In Ranzato, M.; Beygelzimer, A.; Dauphin, Y.; Liang, P.; and Vaughan, J. W., eds., *Advances in Neural Information Processing Systems*, volume 34, 17981–17993. Curran Associates, Inc.

Banarse, D.; Todd, S.; Latham, W.; and Leymarie, F. F. 2026. Evolution & Foundation: AI Shares Creative Control. *arXiv preprint arXiv:2606.16849*.

Black, K.; Janner, M.; Du, Y.; Kostrikov, I.; and Levine, S. 2024. Training Diffusion Models with Reinforcement Learning. In *The Twelfth International Conference on Learning Representations*.

Chess.com. 2020. Forsyth-Edwards Notation (FEN). Blog.

Colton, S.; and Wiggins, G. A. 2012. Computational creativity: The final frontier? In *ECAI 2012 - 20th European Conference on Artificial Intelligence, 27-31 August 2012, Montpellier, France - Including Prestigious Applications of Artificial Intelligence (PAIS-2012) System Demonstration*, Frontiers in Artificial Intelligence and Applications, 21–26. Netherlands: IOS Press. ISBN 9781614990970. European Conference on Artificial Intelligence 2012, ECAI 2012 ; Conference date: 27-08-2012 Through 31-08-2012.

Duplessis, T. 2020. Lichess puzzler. https://github.com/ornicar/lichess-puzzler. Accessed: 2026-1-20.

Duplessis, T. 2026a. Lichess Accuracy Metric. https://lichess.org/page/accuracy. "Accessed: 2026-1-20".

Duplessis, T. 2026b. Lichess puzzle dataset. "Accessed: 2026-1-5".

Earle, S.; Arulkumaran, K.; Dai, A.; Kumar, A.; Togelius, J.; and Risi, S. 2026. In Search of the Ingredients of Open-Endedness: Replicating Picbreeder with Large Vision-Language Models. In *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '26)*. ArXiv preprint arXiv:2605.23908.

Fan, Y.; Watkins, O.; Du, Y.; Liu, H.; Ryu, M.; Boutilier, C.; Abbeel, P.; Ghavamzadeh, M.; Lee, K.; and Lee, K. 2023. DPOK: Reinforcement Learning for Fine-tuning Text-to-Image Diffusion Models. In Oh, A.; Naumann, T.; Globerson, A.; Saenko, K.; Hardt, M.; and Levine, S., eds., *Advances in Neural Information Processing Systems*, volume 36, 79858–79885. Curran Associates, Inc.

Feng, X.; Veeriah, V.; Chiam, M.; Dennis, M. D.; Barbero, F.; Obando-Ceron, J.; Shi, J.; Singh, S.; Hou, S.; Tomasev, N.; and Zahavy, T. 2026. Generating Creative Chess Puzzles. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.

Iqbal, A. 2017. A Computer Composes A Fabled Problem: Four Knights vs. Queen. arXiv:1709.00931.

Iqbal, A. 2021. A Computational Method of Optimizing Chess Compositions to Enhance Aesthetic Appeal. In *2021 2nd International Conference on Artificial Intelligence and Data Sciences (AiDAS)*, 1–6.

Jones, C. R.; and Bergen, B. K. 2026. Large language models pass a standard three-party Turing test. *Proceedings of the National Academy of Sciences*, 123(21): e2524472123.

Kool, W.; van Hoof, H.; and Welling, M. 2019. Buy 4 REINFORCE Samples, Get a Baseline for Free! In *International Conference on Learning Representations*.

Luo, X.; Yu, W.; Li, Q.; and Bjerva, J. 2026. Characterizing Memorization in Diffusion Language Models: Generalized Extraction and Sampling Effects. In *ICML 2026 Workshop on Foundations of Deep Generative Models: Understanding Memorization, Generalization, and Reasoning*.

Nie, S.; Zhu, F.; Du, C.; Pang, T.; Liu, Q.; Zeng, G.; Lin, M.; and Li, C. 2025a. Scaling up Masked Diffusion Models on Text. In *The Thirteenth International Conference on Learning Representations*.

Nie, S.; Zhu, F.; You, Z.; Zhang, X.; Ou, J.; Hu, J.; Zhou, J.; Lin, Y.; Wen, J.-R.; and LI, C. 2025b. Large Language Diffusion Models. In Belgrave, D.; Zhang, C.; Lin, H.; Pascanu, R.; Koniusz, P.; Ghassemi, M.; and Chen, N., eds., *Advances in Neural Information Processing Systems*, volume 38, 50608–50646. Curran Associates, Inc.

Ou, J.; Han, J.; Xu, M.; Xu, S.; Xie, J.; Ermon, S.; Wu, Y.; and Li, C. 2026. Principled RL for Diffusion LLMs Emerges from a Sequence-Level Perspective. In *The Fourteenth International Conference on Learning Representations*.

Rojas, K.; Lin, J.; Rasul, K.; Schneider, A.; Nevmyvaka, Y.; Tao, M.; and Deng, W. 2026. Improving Reasoning for Diffusion Language Models via Group Diffusion Policy Optimization. In *The Fourteenth International Conference on Learning Representations*.

Ruoss, A.; Delétang, G.; Medapati, S.; Grau-Moya, J.; Wenliang, L. K.; Catt, E.; Reid, J.; Lewis, C. A.; Veness, J.; and Genewein, T. 2024. Amortized Planning with Large-Scale Transformers: A Case Study on Chess. In Globerson, A.; Mackey, L.; Belgrave, D.; Fan, A.; Paquet, U.; Tomczak, J.; and Zhang, C., eds., *Advances in Neural Information Processing Systems*, volume 37, 65765–65790. Curran Associates, Inc.

Schulman, J.; Wolski, F.; Dhariwal, P.; Radford, A.; and Klimov, O. 2017. Proximal Policy Optimization Algorithms. arXiv:1707.06347.

Shao, Z.; Wang, P.; Zhu, Q.; Xu, R.; Song, J.; Bi, X.; Zhang, H.; Zhang, M.; Li, Y. K.; Wu, Y.; and Guo, D. 2024. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. arXiv:2402.03300.

Shi, J.; Han, K.; Wang, Z.; Doucet, A.; and Titsias, M. 2024. Simplified and Generalized Masked Diffusion for Discrete Data. In Globerson, A.; Mackey, L.; Belgrave, D.; Fan, A.; Paquet, U.; Tomczak, J.; and Zhang, C., eds., *Advances in Neural Information Processing Systems*, volume 37, 103131–103167. Curran Associates, Inc.

Turing, A. M. 1950. Computing machinery and intelligence. *Mind*, 59(236): 433–460.

Veeriah, V.; Barbero, F.; Chiam, M.; Feng, X.; Dennis, M.; Pachauri, R.; Tumiel, T.; Obando-Ceron, J.; Shi, J.; Hou, S.; Singh, S.; Tomašev, N.; and Zahavy, T. 2025. Evaluating In Silico Creativity: An Expert Review of AI Chess Compositions. In *NeurIPS 2025 Creative AI Track*.

Wang, G.; Turok, G.; Schiff, Y.; Arriola, M.; and Kuleshov, V. 2026. d2: Improving Reasoning in Diffusion Language Models via Trajectory Likelihood Estimation. In *Forty-third International Conference on Machine Learning*.

Wenger, E.; and Kenett, Y. N. 2026. Large language models are homogeneously creative. *PNAS Nexus*, 5(3): pgag042.

Wu, C.; Zhang, H.; Xue, S.; Liu, Z.; Diao, S.; Zhu, L.; Luo, P.; Han, S.; and Xie, E. 2026. Fast-dLLM: Training-free Acceleration of Diffusion LLM by Enabling KV Cache and Parallel Decoding. In *The Fourteenth International Conference on Learning Representations*.

Ye, J.; Gao, J.; Gong, S.; Zheng, L.; Jiang, X.; Li, Z.; and Kong, L. 2025a. Beyond Autoregression: Discrete Diffusion for Complex Reasoning and Planning. In Yue, Y.; Garg, A.; Peng, N.; Sha, F.; and Yu, R., eds., *International Conference on Learning Representations*, volume 2025, 77875–77898.

Ye, J.; Xie, Z.; Zheng, L.; Gao, J.; Wu, Z.; Jiang, X.; Li, Z.; and Kong, L. 2025b. Dream 7B: Diffusion Large Language Models. arXiv:2508.15487.

Yujian, L.; and Bo, L. 2007. A Normalized Levenshtein Distance Metric. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 29(6): 1091–1095.

Zahavy, T.; Hou, S.; et al. 2026. COrigami: An AI Pipeline for Co-Designing Flat-Foldable Visually Recognisable Origami. *arXiv preprint arXiv:2606.26299*.

Zhao, S.; Gupta, D.; Zheng, Q.; and Grover, A. 2026. d1: Scaling Reasoning in Diffusion Large Language Models via Reinforcement Learning. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.